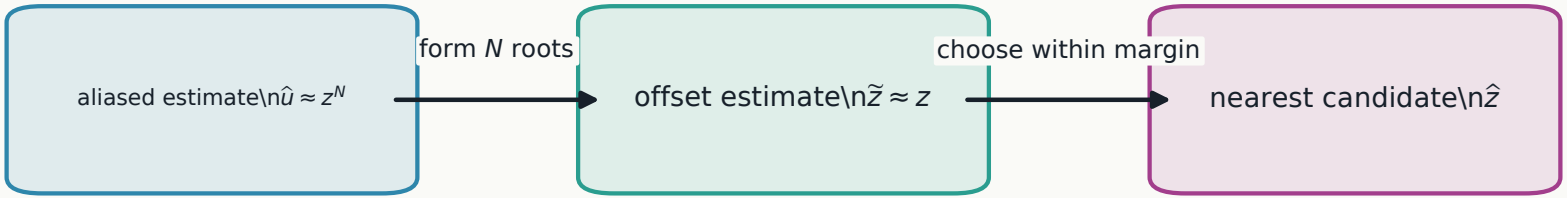
Branch selection is certified by a geometric margin



$$\text{coarse phase error} + \text{root perturbation} < \sin(\pi/N)$$